



Computer Network

Chapter 1: Introduction Network Overview



Content



-
- Exploring the functions of networking
 - Securing the network
 - OSI model

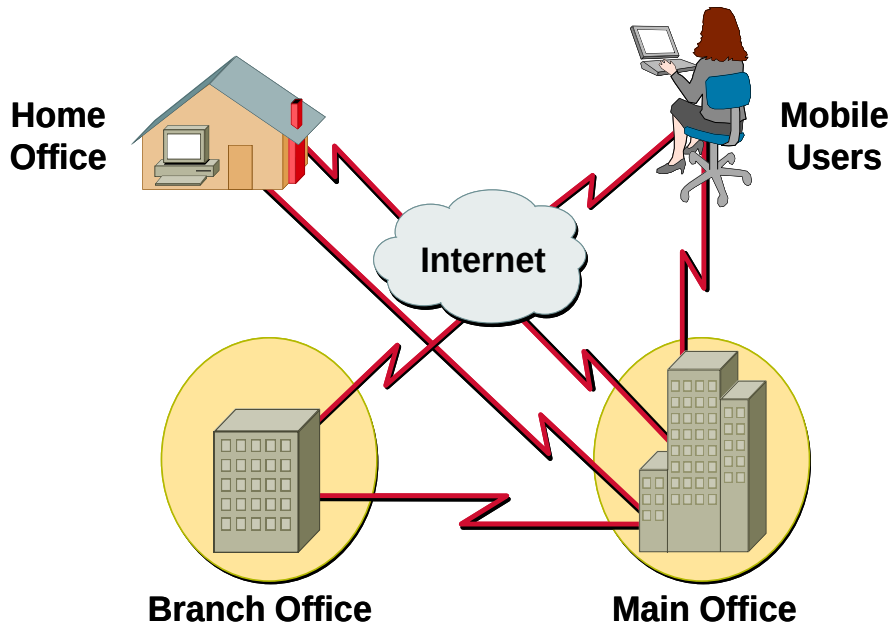


What is a network?

- A network consists of two or more computers that are linked in order to share resources (such as printers and CDs), exchange files, or allow electronic communications. The computers on a network may be linked through cables, telephone lines, radio waves, satellites, or infrared light beams.
- Environments
 - Homes
 - Small businesses
 - Large enterprises



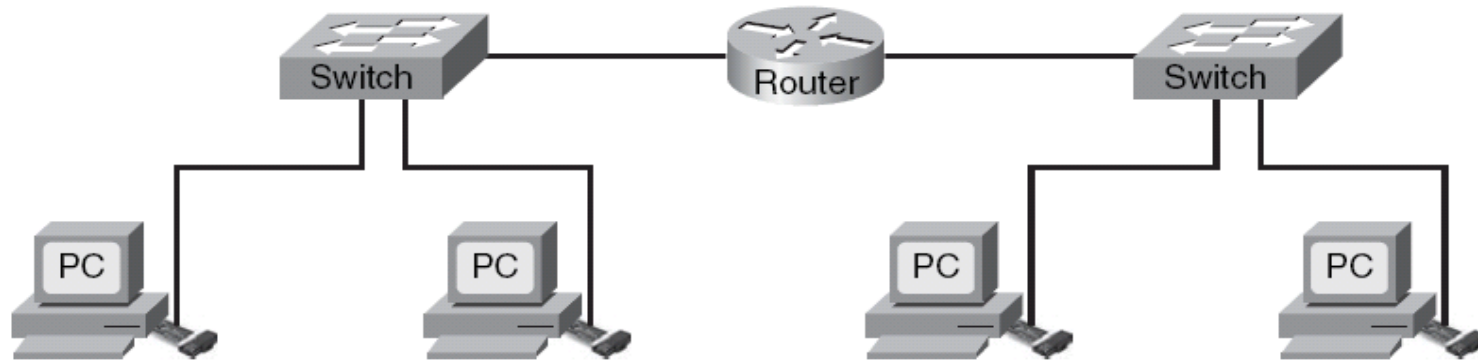
Large enterprise network



- Main office
 - Several buildings
 - Hundreds of employee
- Remote location
 - Branch office:
 - Local network resource
 - Access information from main office
 - Home office
 - on-demand connections to the main or branch offices
 - Mobile users
 - Connect while traveling



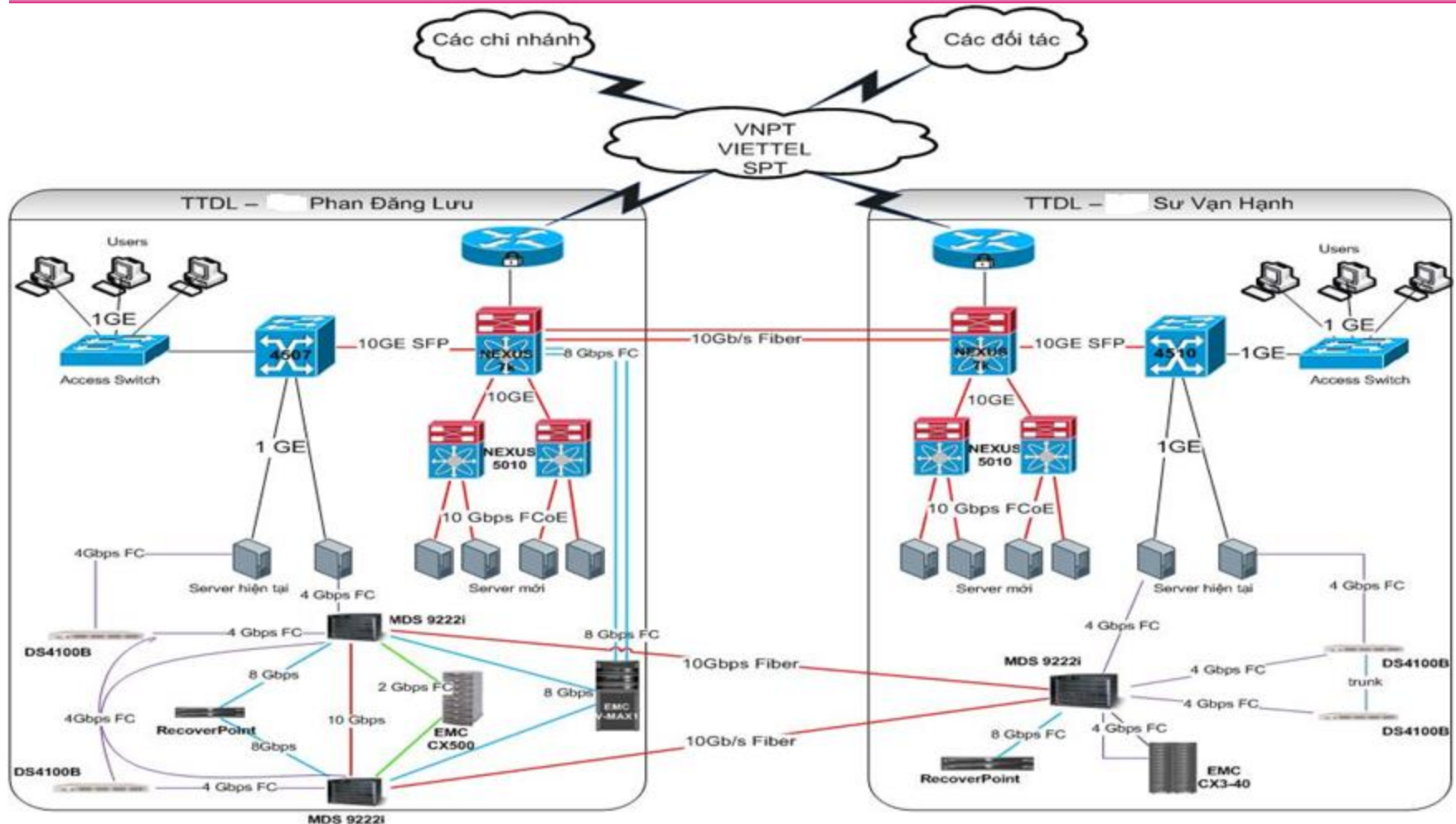
Network devices



- PCs:
 - End devices
 - Data source & destination
- Connection Link
 - NIC
 - Wire: cable; wireless
 - Cable pin
- Switch:
 - Connect end-devices
 - Transmit data in local network
- Routers
 - Connect different networks
 - Routing

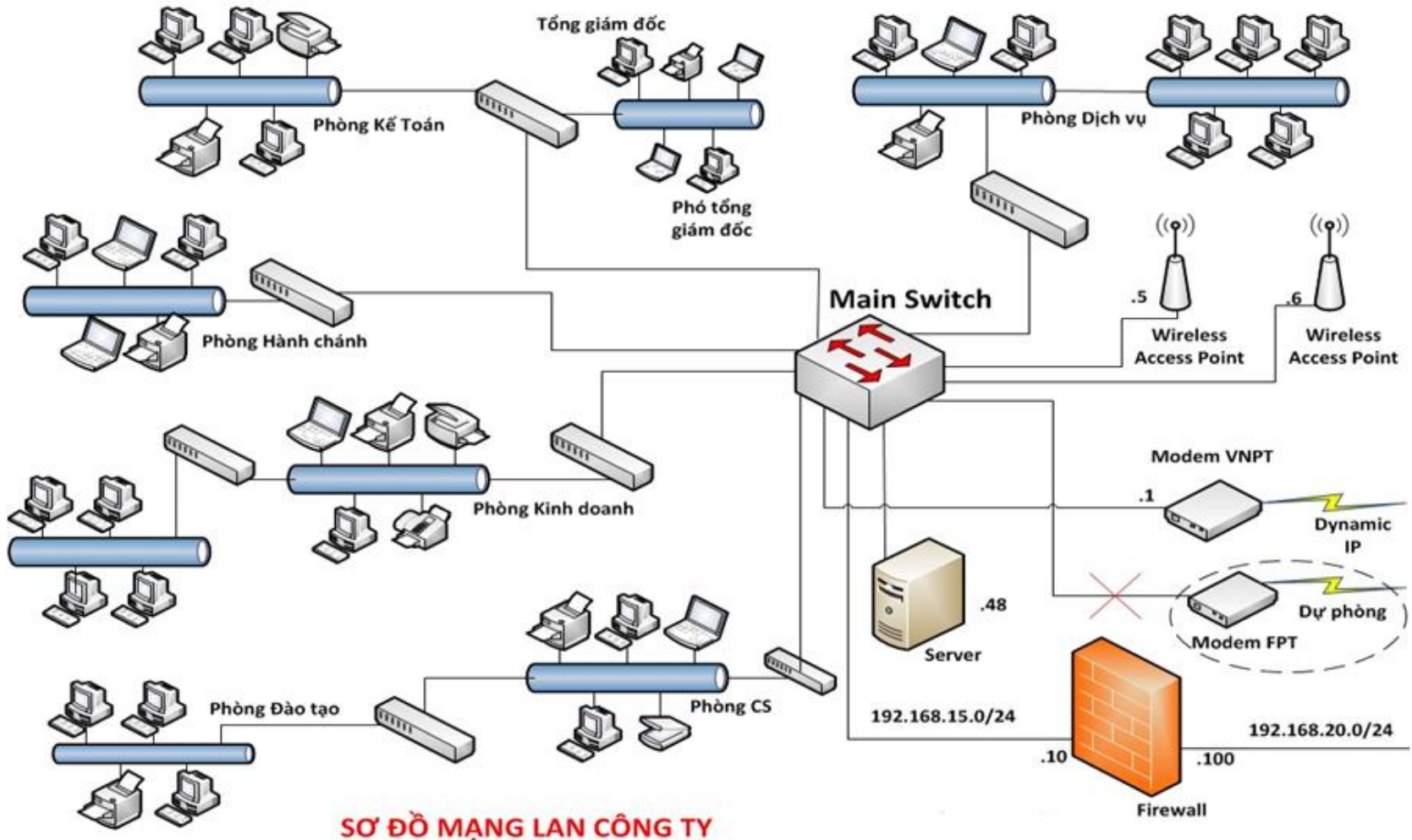


Connection Internet Style





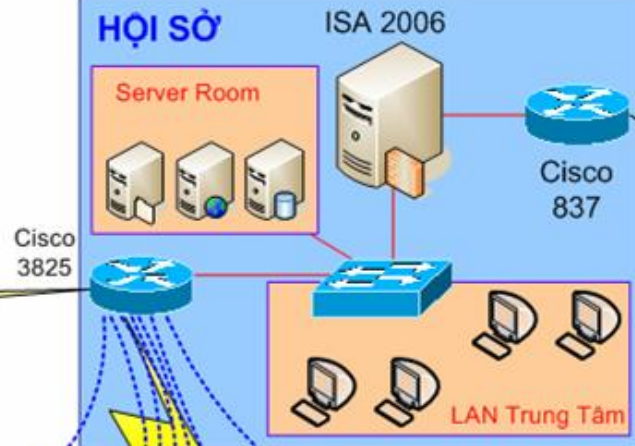
Connection Internet Style



Chi nhánh Bình Dương



HỘI SỞ



Leased line 1Mbps

PVC 128Kbps

PVC 128Kbps

PVC 128Kbps

PVC 128Kbps

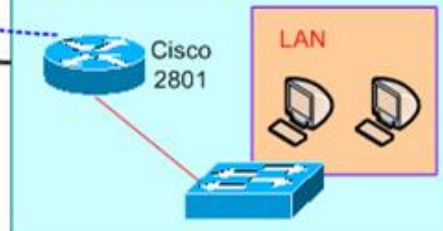
PVC 128Kbps

PVC 128Kbps

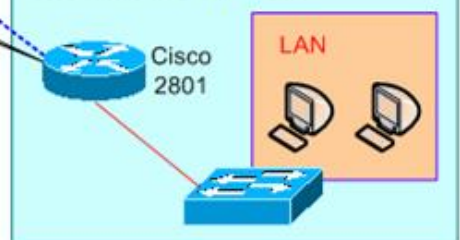
PVC 128Kbps

Frame Relay

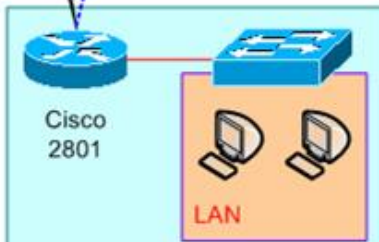
Chi nhánh Sài Gòn



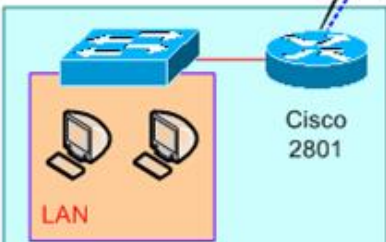
Chi nhánh Chợ Lớn



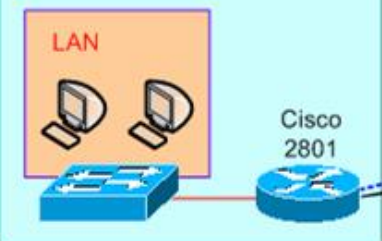
Chi nhánh Hoàng Kiếm



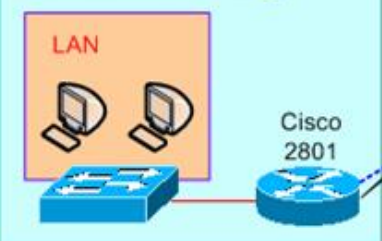
Chi nhánh Hà Nội



Chi nhánh Đồng Nai



Chi nhánh Đà Nẵng





Network Security

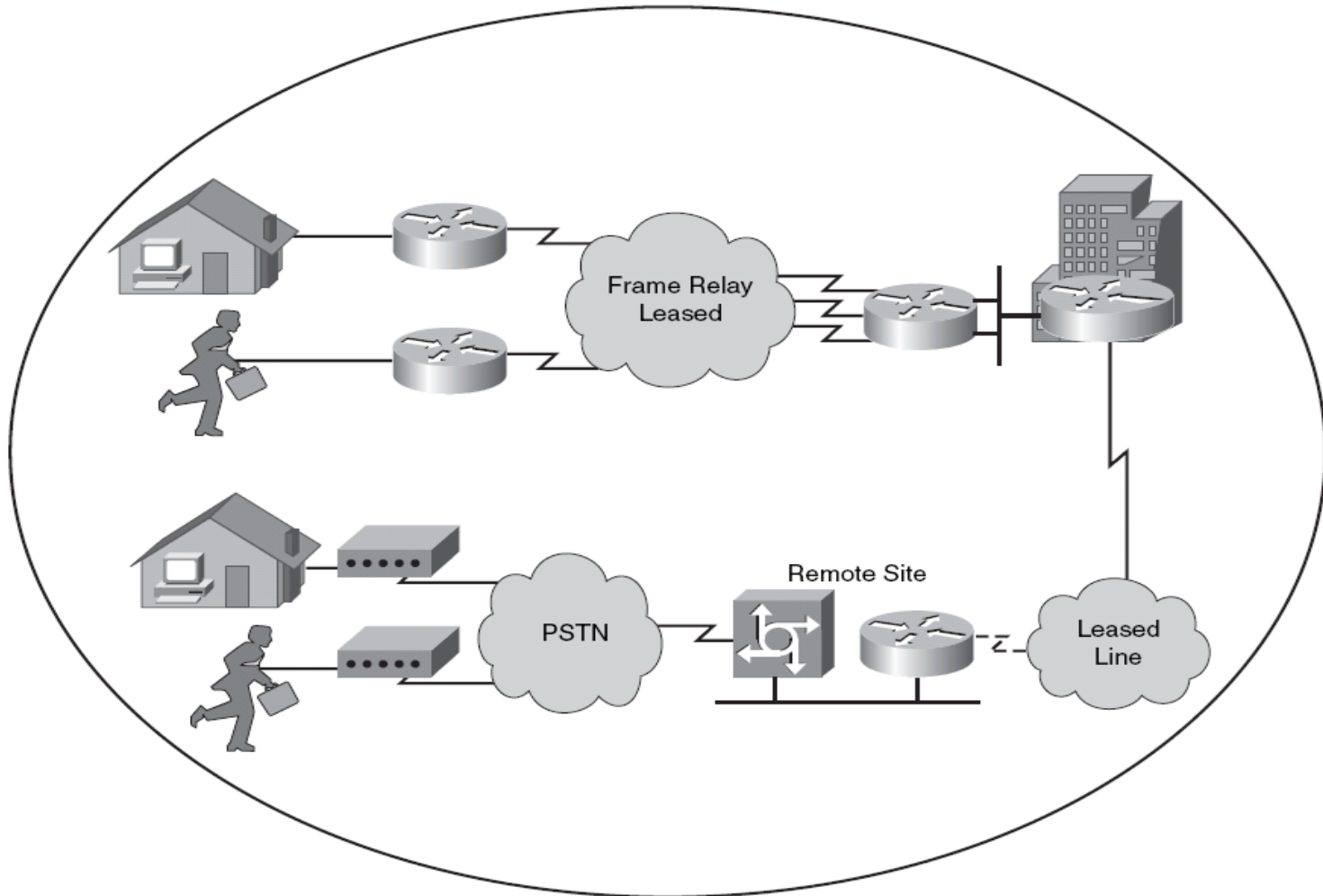


- Closed network
 - Secure from outside
 - Misuse from inside
- Open network
 - More threats
 - Grow e-business
- Balance between Closed network and Open network





Closed network







Devices symbols



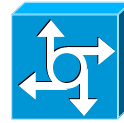
Bridge



Switch



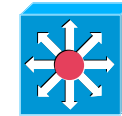
Router



**Access
server**



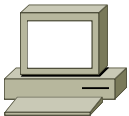
**ISDN
switch**



**Multi-layer
switch**



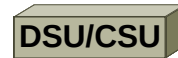
**Network
switch**



**Personal
computer**



File Server



**Data Service Unit/
Channel Service Unit**



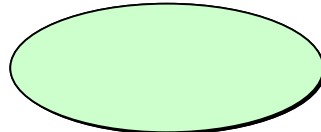
Modem



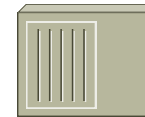
Web Server



WAN "cloud"



**VLAN
(Color May Vary)**



Hub



**Network Cloud or
Broadcast Domain**



Ethernet



Fast Ethernet



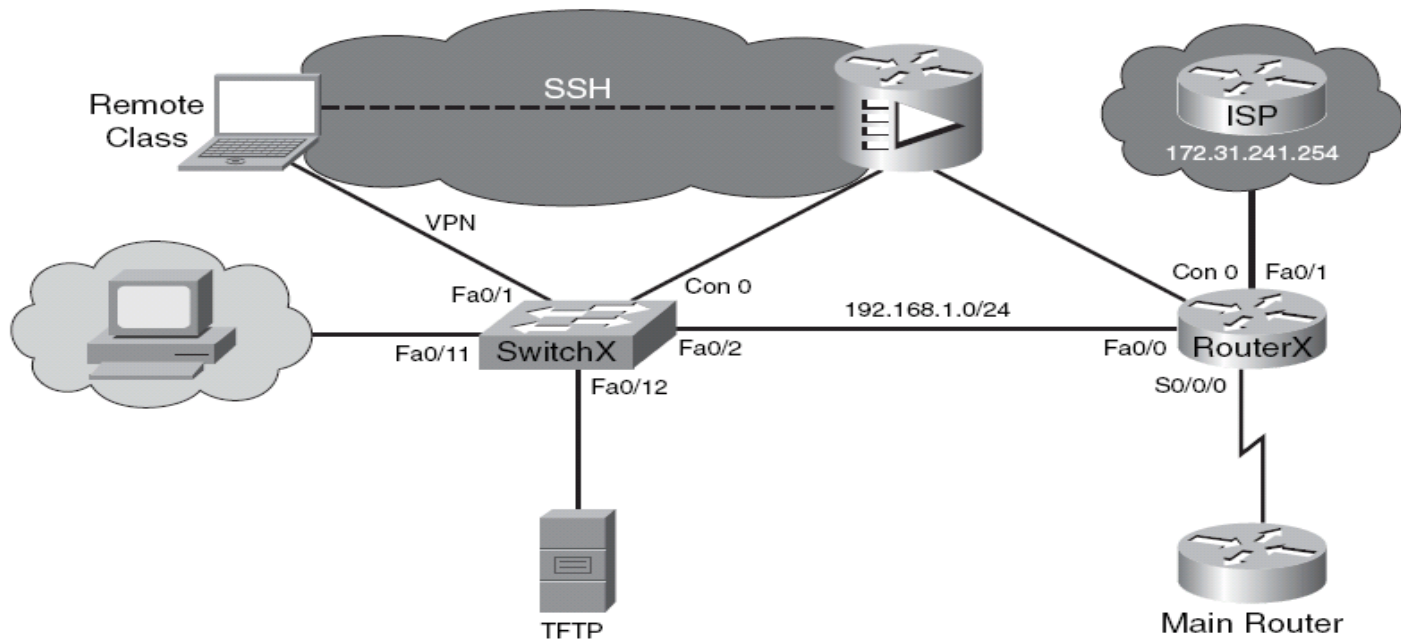
Serial Line



**Circuit Switched
Line**



Mapping



- Cloud: Internet or WAN
- Round Shape: router.
- Square Shape: switch.
- PC: server.
- Laptop /monitor: End-devices
- Straight Line: Ethernet connection
- Z line: serial connection



Characteristics of a Network

- Speed (data rate):
 - How fast data is transmitted
- Cost
 - Cost of components, installation, and maintenance
- Security
 - Both storage data and transmitted data
- Availability
 - 24/7
- Scalability
 - Meet new needs when the network grows.
- Reliability
 - Probability of failure
- Topology



Reduce common threats



- Physical installation
 - Room: locked, only 1 door, electronic access, security cameras,...
 - Environment: temperature and humidity control systems; electrostatic and magnetic interference; remotely monitor and alarm the environmental; ...
 - Electrical: UPS, ...
 - Maintenance: label all equipment cabling; rack-to-rack connections; do not leave a console connected; always log off administrative interfaces
- Reconnaissance Attacks
 - information gathering
 - determine which IP addresses
 - determines which services or ports are active
 - determine the type and version of the application and operating system
- Password Attacks
 - Do not allow users to have the same password on multiple systems.
 - Disable accounts after a specific number of unsuccessful logins.
 - Do not use plaintext passwords.
 - Use strong passwords.

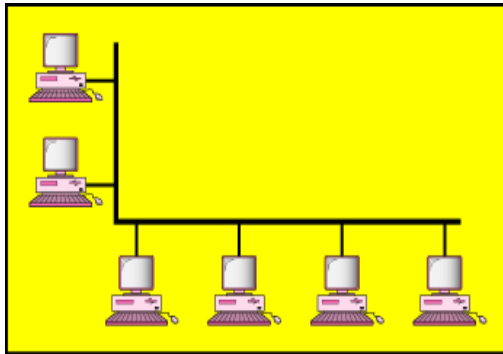


Classification of computer networks

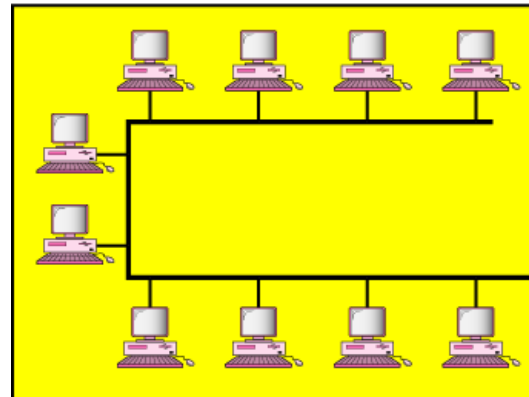
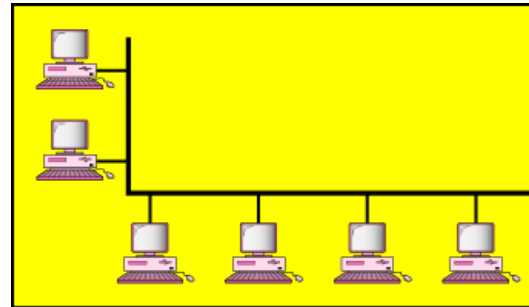
- Environments
 - LAN
 - MAN
 - WAN
- Design
 - Workgroup (Peer-to-Peer)
 - Client/Server
- Topology
 - Bus
 - Star
 - Ring
 - Mesh
 -



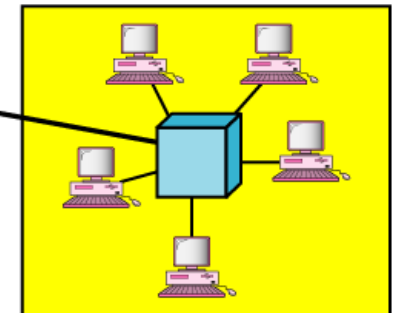
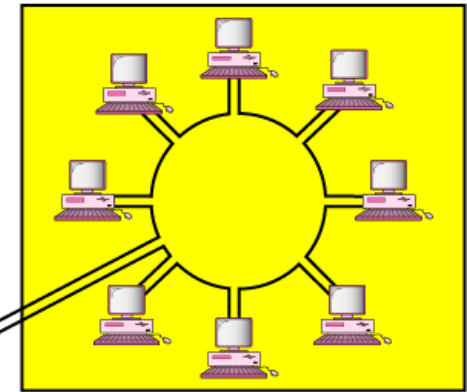
LAN models



a. Single-building LAN



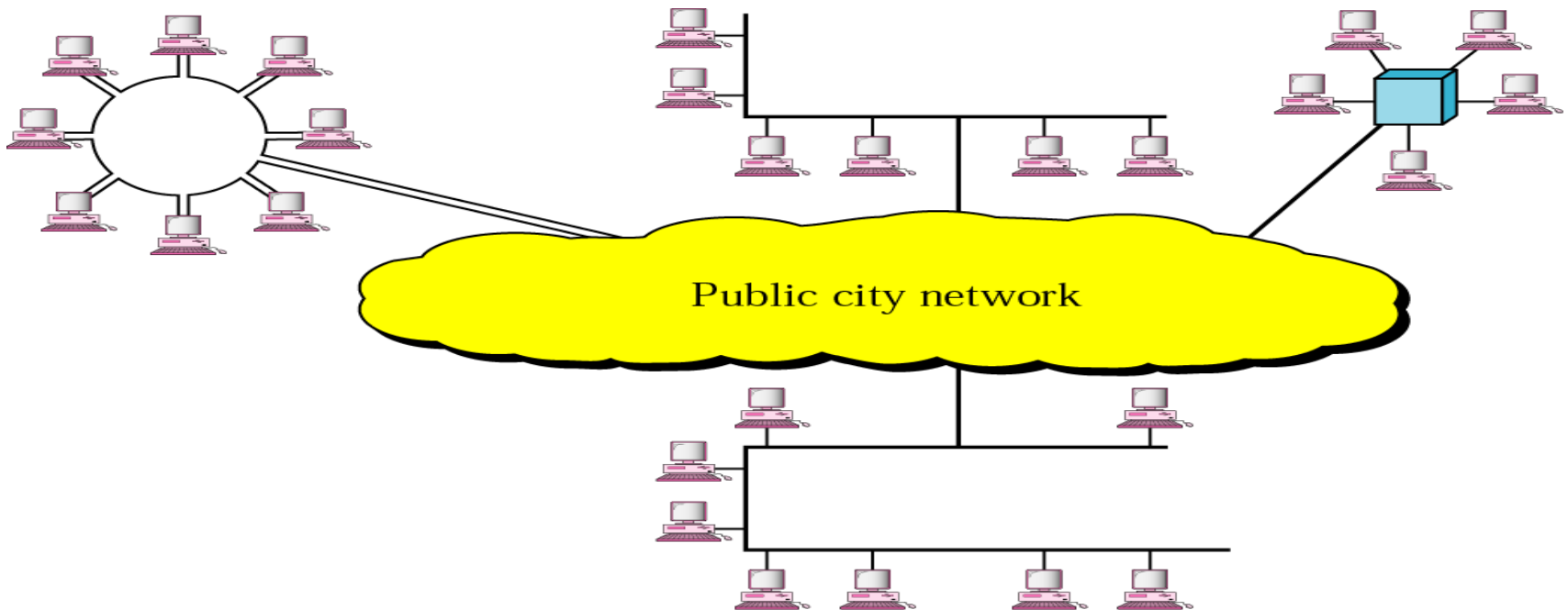
Backbone



b. Multiple-building LAN



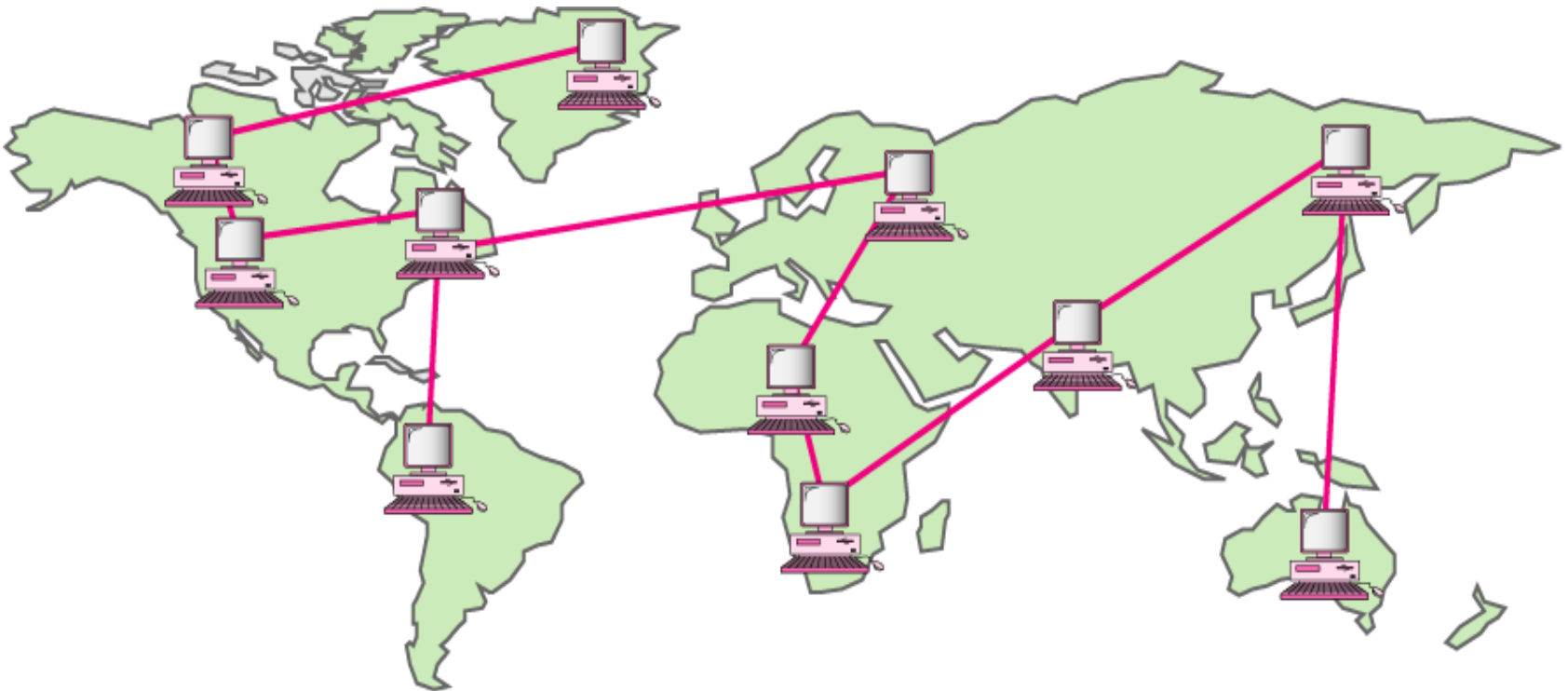
Metropolitan Area Networks (MANs)



- Cover a city
- Interconnecting networks for a big company in a metropolitan area
- Owned by Network Operators or big companies
- Exp, Cable TV networks



Wide Area Networks (WANs)



- Cover a country or continent



Network Protocol

- A **network protocol** defines rules and conventions for communication between network devices.
- **Network protocols** include mechanisms for devices to identify and make connections with each other, as well as formatting rules that specify how data is packaged into messages sent and received

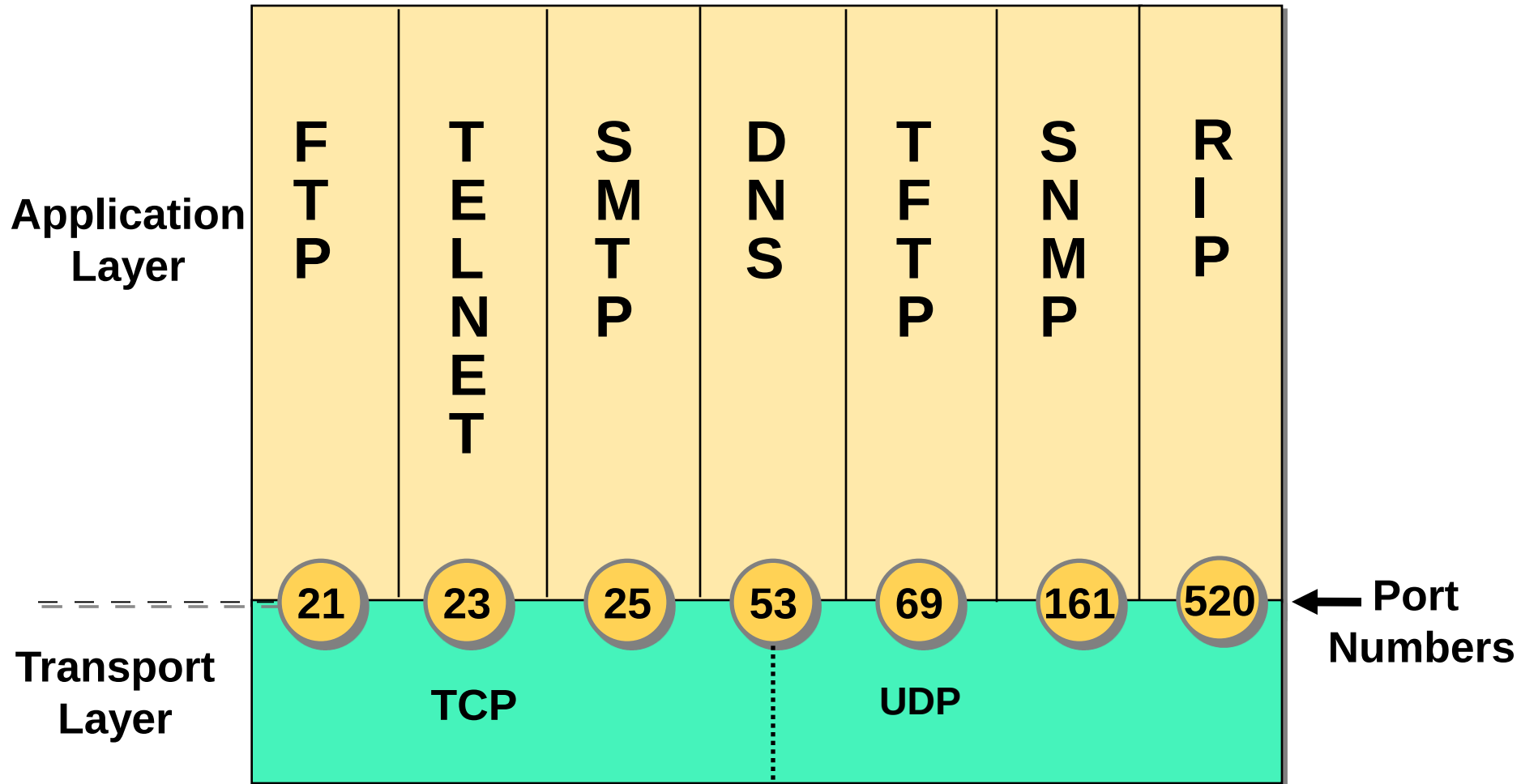


Network Protocols

- TCP
- UDP
- FTP
- HTTP
- HTTPS
- DNS
- SMTP
- POP
- Telnet
- SSH
-



Port Numbers





Protocol vs Standard

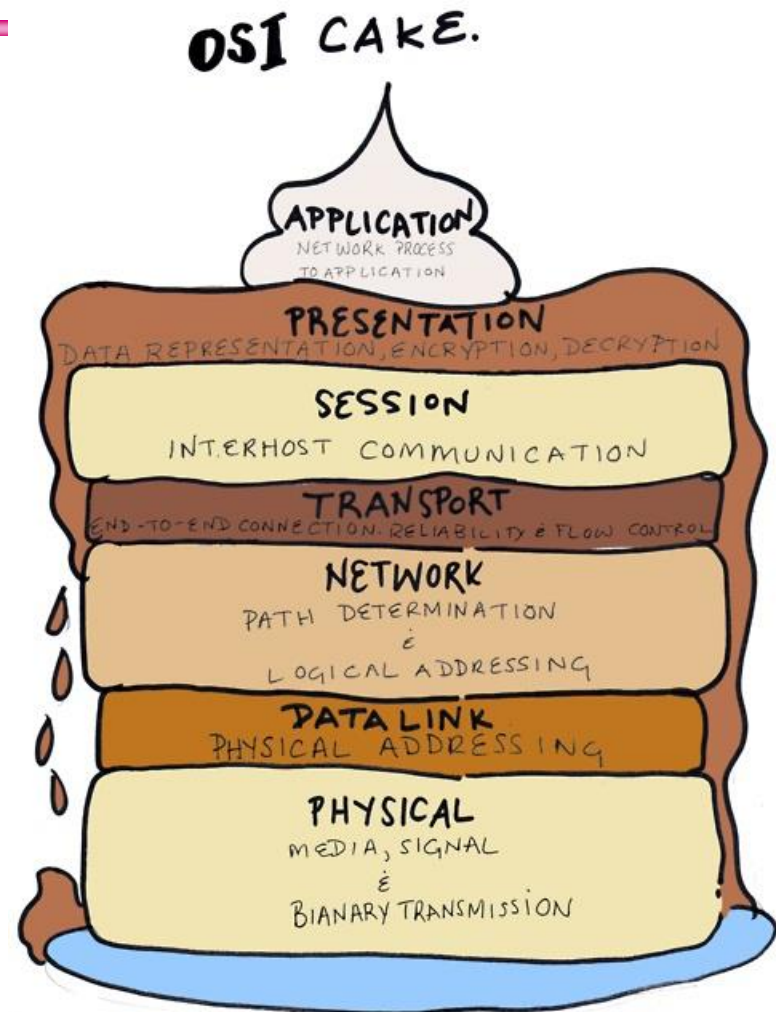
- **Protocol**: a set of rules that governs data communication.
- **Standards**: agreed-upon rules.



OSI Introduction



- Multi-vendors
- → one standard
- → OSI model





OSI Introduction



-
- Established in 1947, the ***International Standards Organization (ISO)*** is a multinational body dedicated to worldwide agreement on international standards. Almost three-fourths of countries in the world are represented in the ISO.
 - An ISO standard that covers all aspects of network communications is the **Open Systems Interconnection (OSI)** model. It was first introduced in the late 1970s.



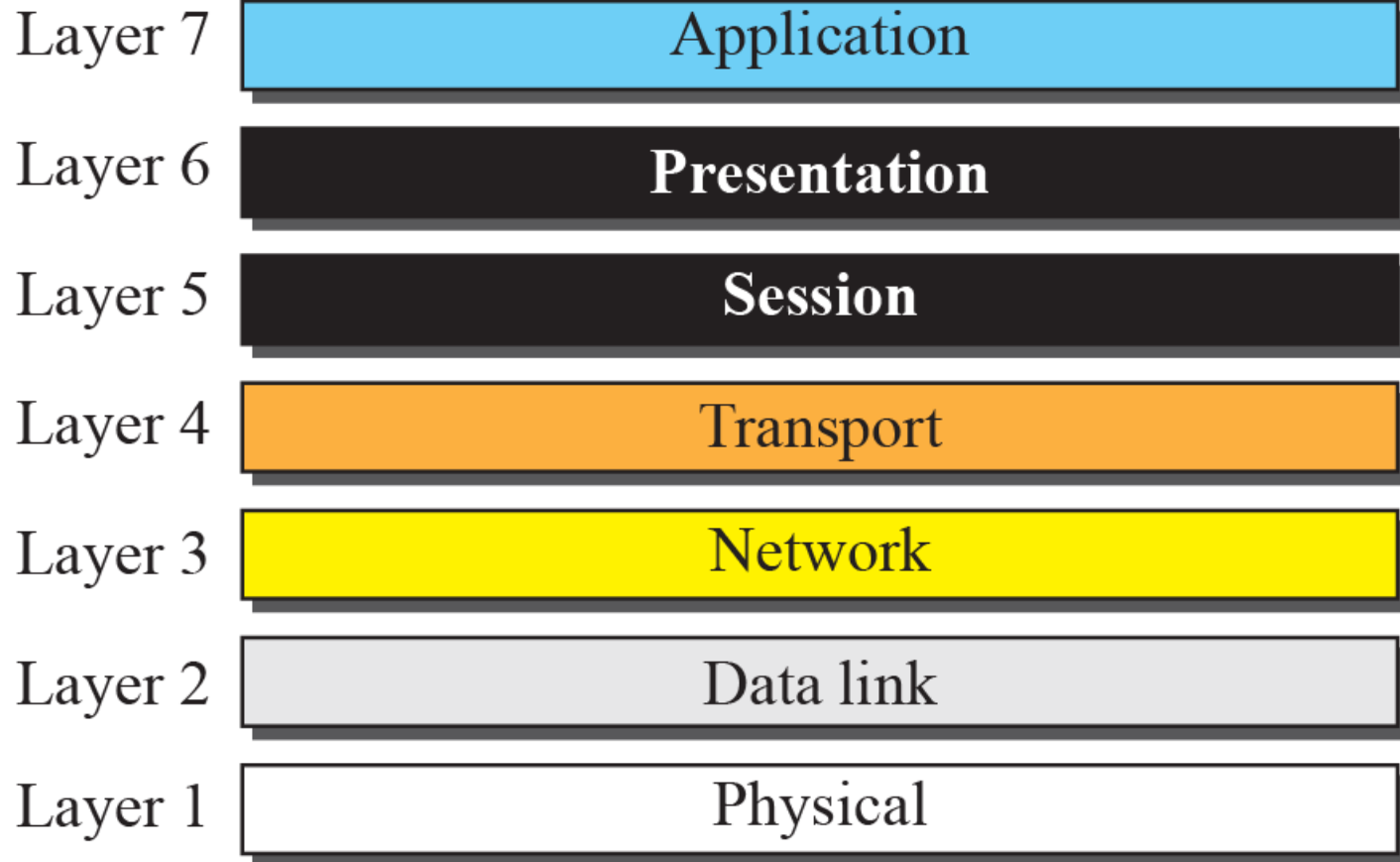
OSI Introduction

- **OSI:** is a set of protocols that allow any two different systems to communicate regardless of their underlying architecture. Allows complete interoperability between systems

*ISO is the organization;
OSI is the model.*

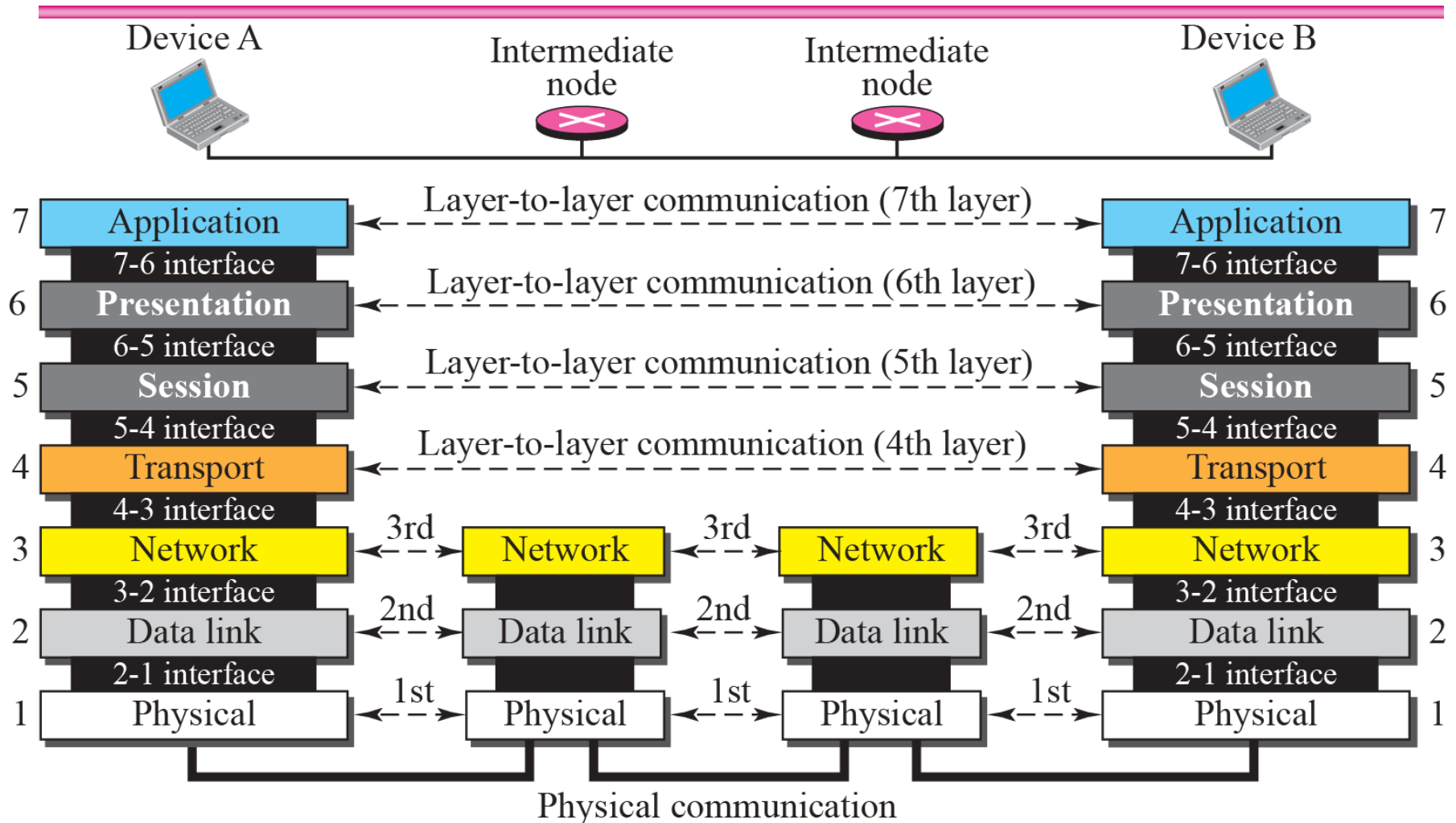


OSI model overview





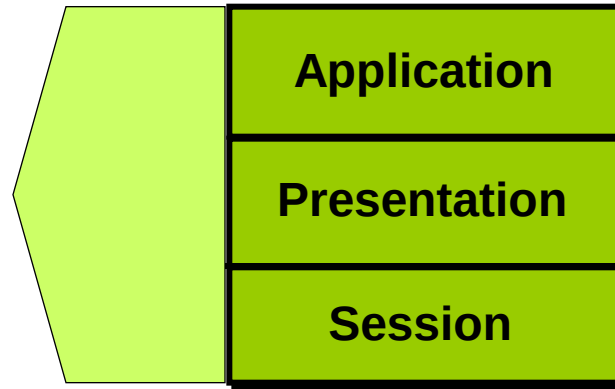
OSI model overview





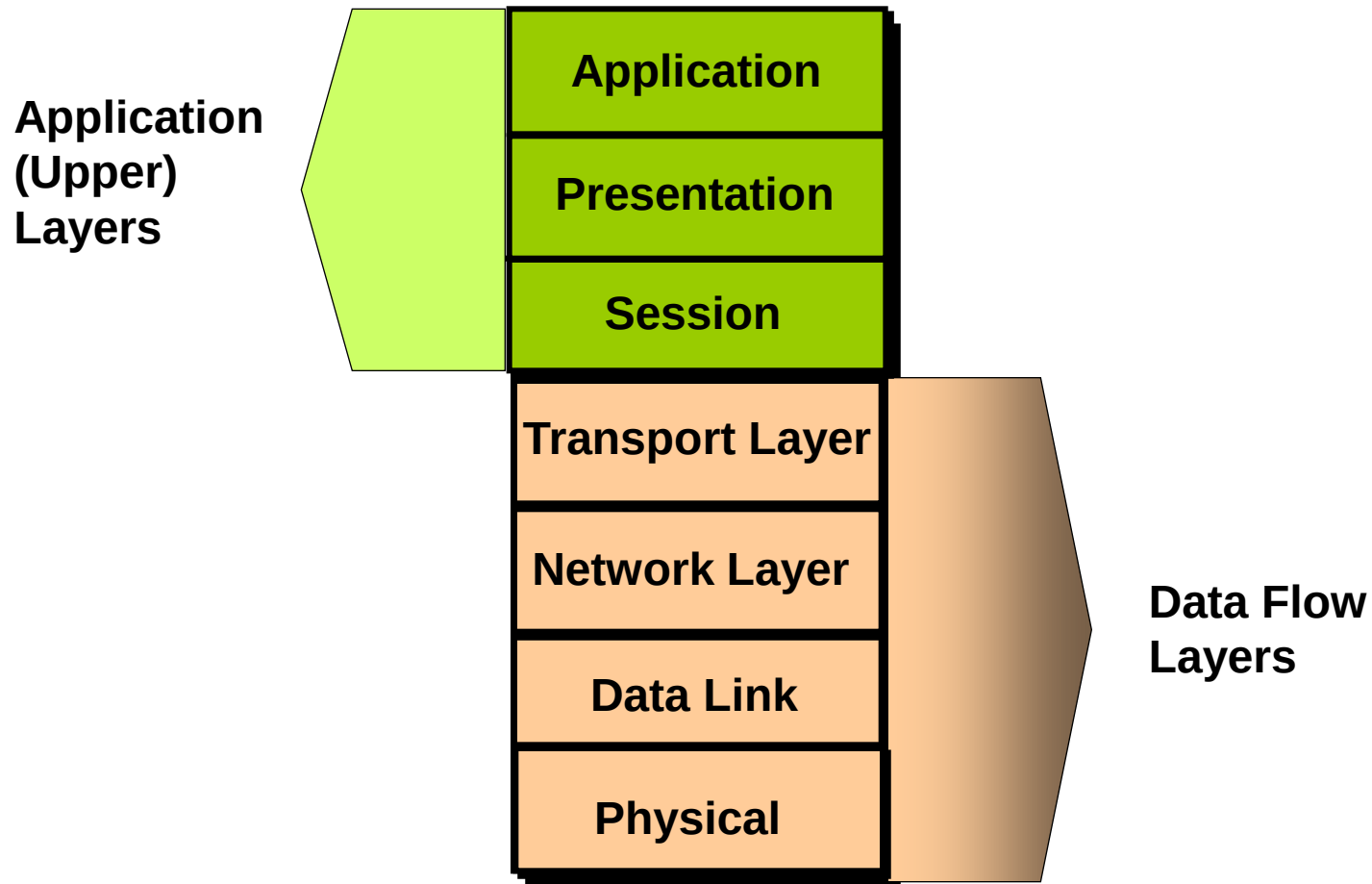
OSI model overview

**Application
(Upper)
Layers**





OSI model overview





Role of Application Layers



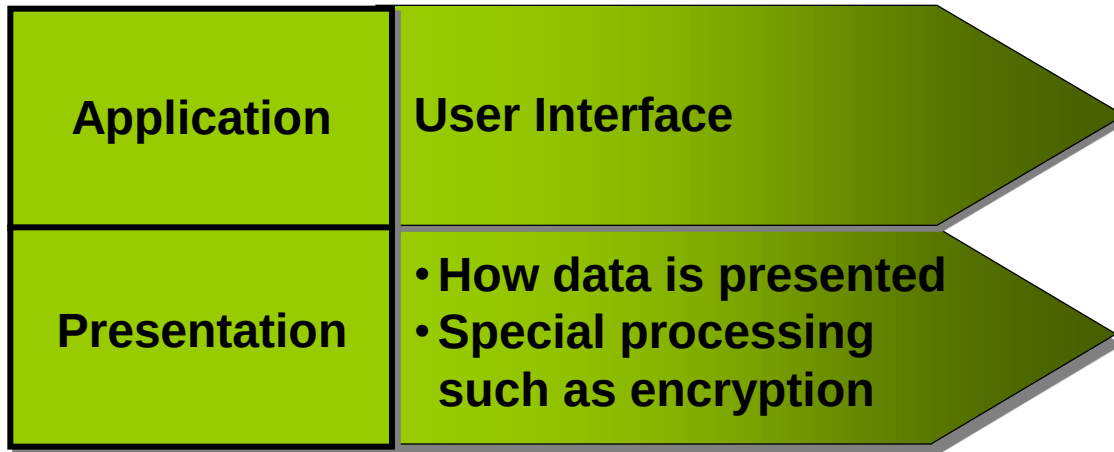
EXAMPLES



Telnet
FTP



Role of Application Layers



EXAMPLES

Telnet
FTP

ASCII
EBCDIC
JPEG



Role of Application Layers

Application	User Interface
Presentation	• How data is presented • Special processing such as encryption
Session	Keeping different applications' data separate

EXAMPLES

Telnet
FTP

ASCII
EBCDIC
JPEG

Operating System/
Application Access
Scheduling



Role of Application Layers

EXAMPLES

Application	User Interface	Telnet FTP
Presentation	• How data is presented • Special processing such as encryption	ASCII EBCDIC JPEG
Session	Keeping different applications' data separate	Operating System/ Application Access Scheduling
Transport Layer		
Network Layer		
Data Link		
Physical		



Role of Data Flow Layers

EXAMPLES

Physical	• Move bits between devices • Specifies voltage, wire speed and pin-out cables	EIA/TIA-232 V.35
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Role of Data Flow Layers

EXAMPLES

Data Link	• Combines bits into bytes and bytes into frames • Access to media using MAC address • Error detection not correction	802.3 / 802.2 HDLC
Physical	• Move bits between devices • Specifies voltage, wire speed and pin-out cables	EIA/TIA-232 V.35



Role of Data Flow Layers

EXAMPLES

Network	Provide logical addressing which routers use for path determination	IP IPX
Data Link	• Combines bits into bytes and bytes into frames • Access to media using MAC address • Error detection not correction	802.3 / 802.2 HDLC
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Role of Data Flow Layers

EXAMPLES

Transport	• Reliable or unreliable delivery • Error correction before retransmit	TCP UDP SPX
Network	• Provide logical addressing which routers use for path determination	IP IPX
Data Link	• Combines bits into bytes and bytes into frames • Access to media using MAC address • Error detection (not correction)	802.3 / 802.2 HDLC
Physical	• Move bits between devices • Specifies voltage, wire speed and pin-out cables	EIA/TIA-232 V.35



Role of Data Flow Layers

Application		
Presentation		
Session		<u>EXAMPLES</u>
Transport	• Reliable or unreliable delivery • Error correction before retransmit	TCP UDP SPX
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Physical	• Move bits between devices • Specifies voltage, wire speed and pin-out cables	EIA/TIA-232 V.35



OSI model overview

OSI model:

Upper layers (L4 – L7): implemented mostly in software

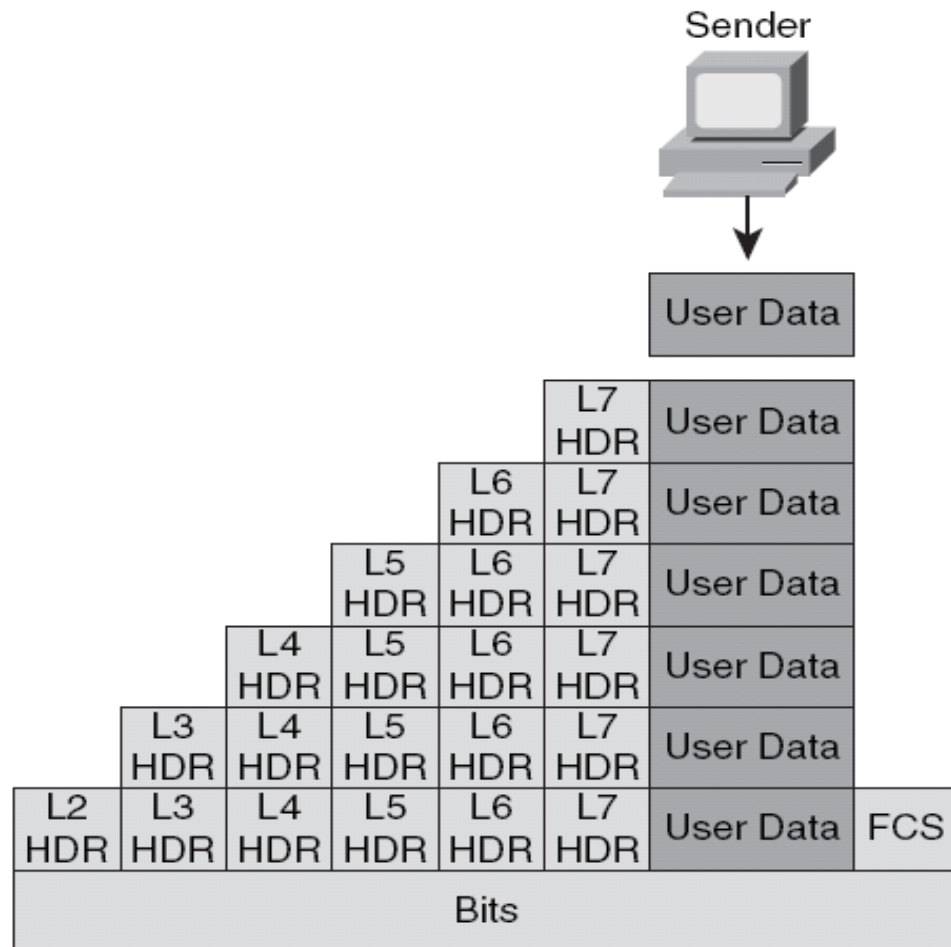
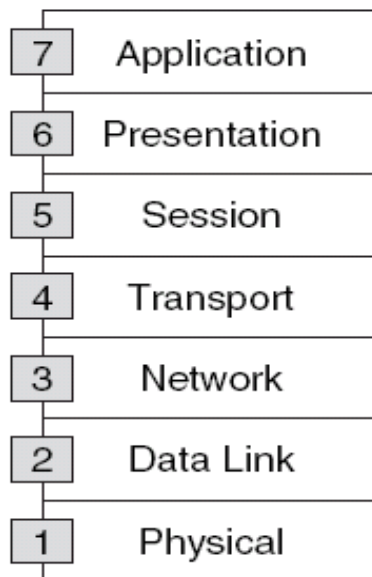
Lower layers (L2 & L3): combination of software and hardware.

Physical layer (L1): mostly in hardware.

Each layer adds **header** to the message except layer 2 adds header and a **trailer**.



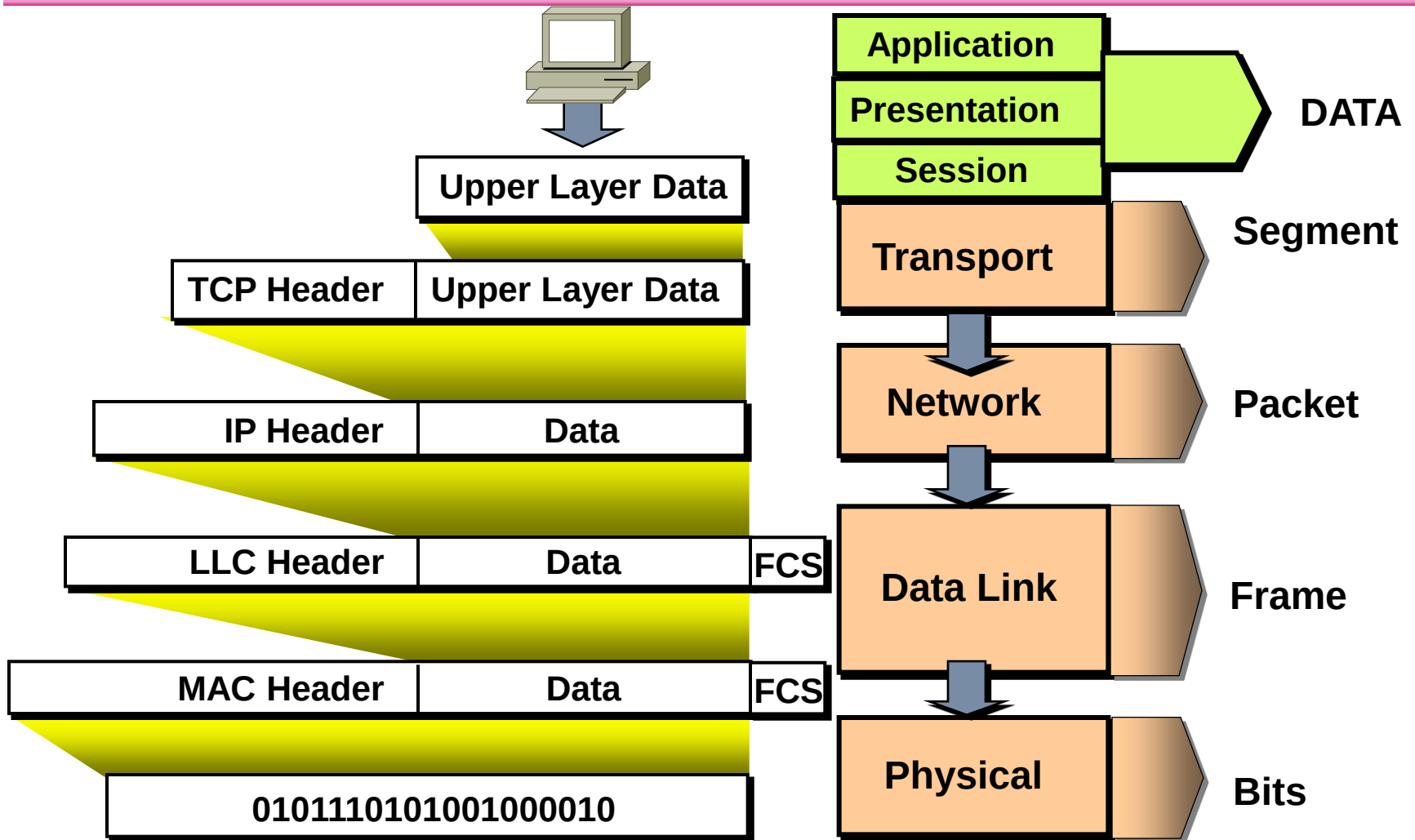
Encapsulating Data



HDR = Header

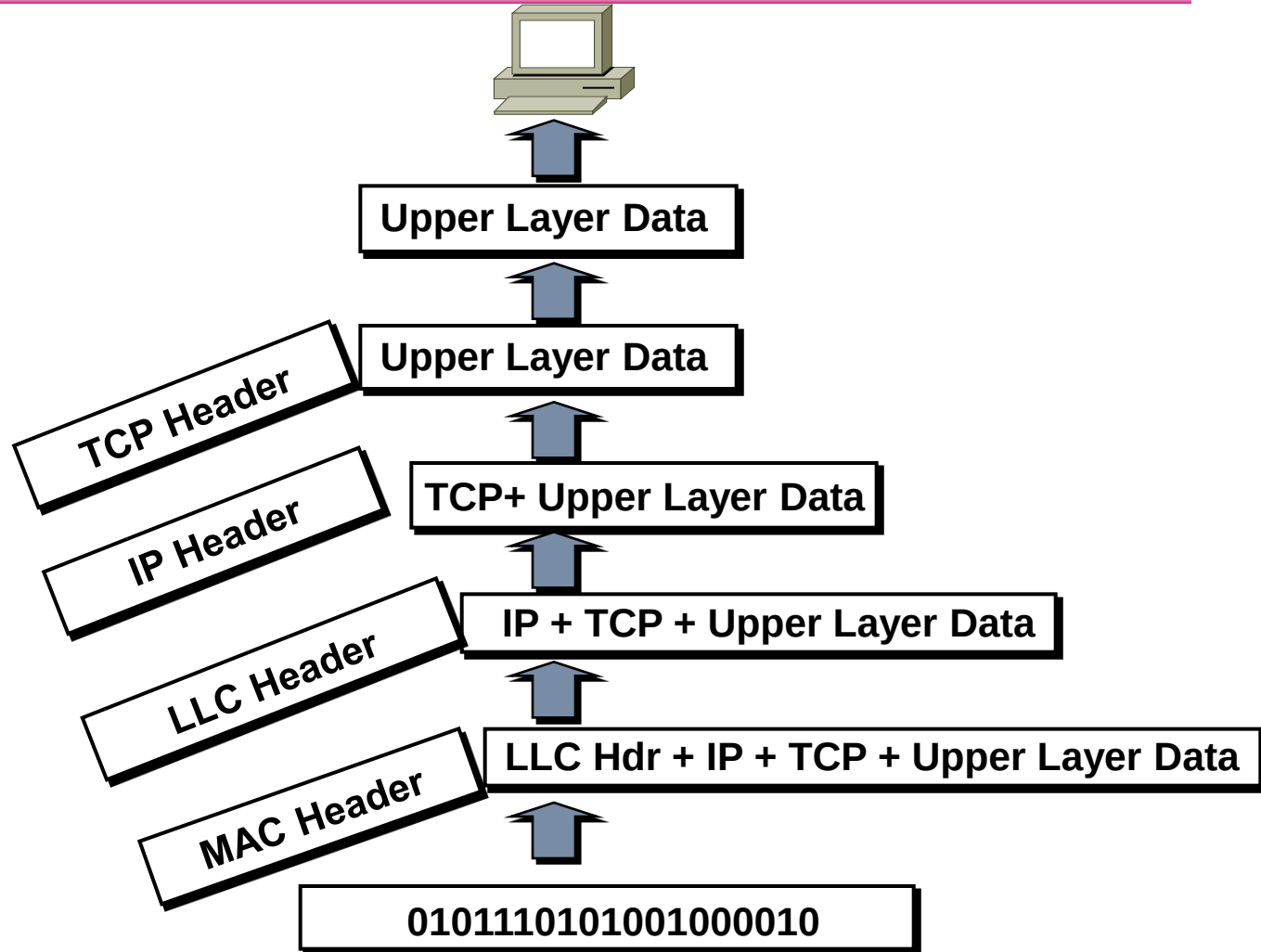
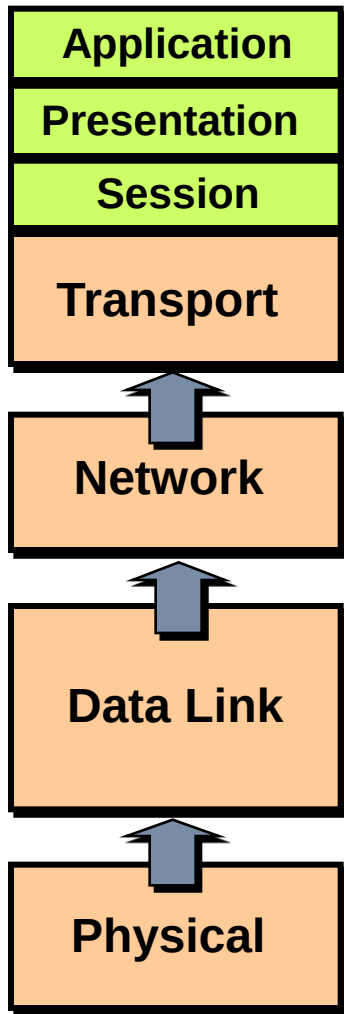


Encapsulating Data



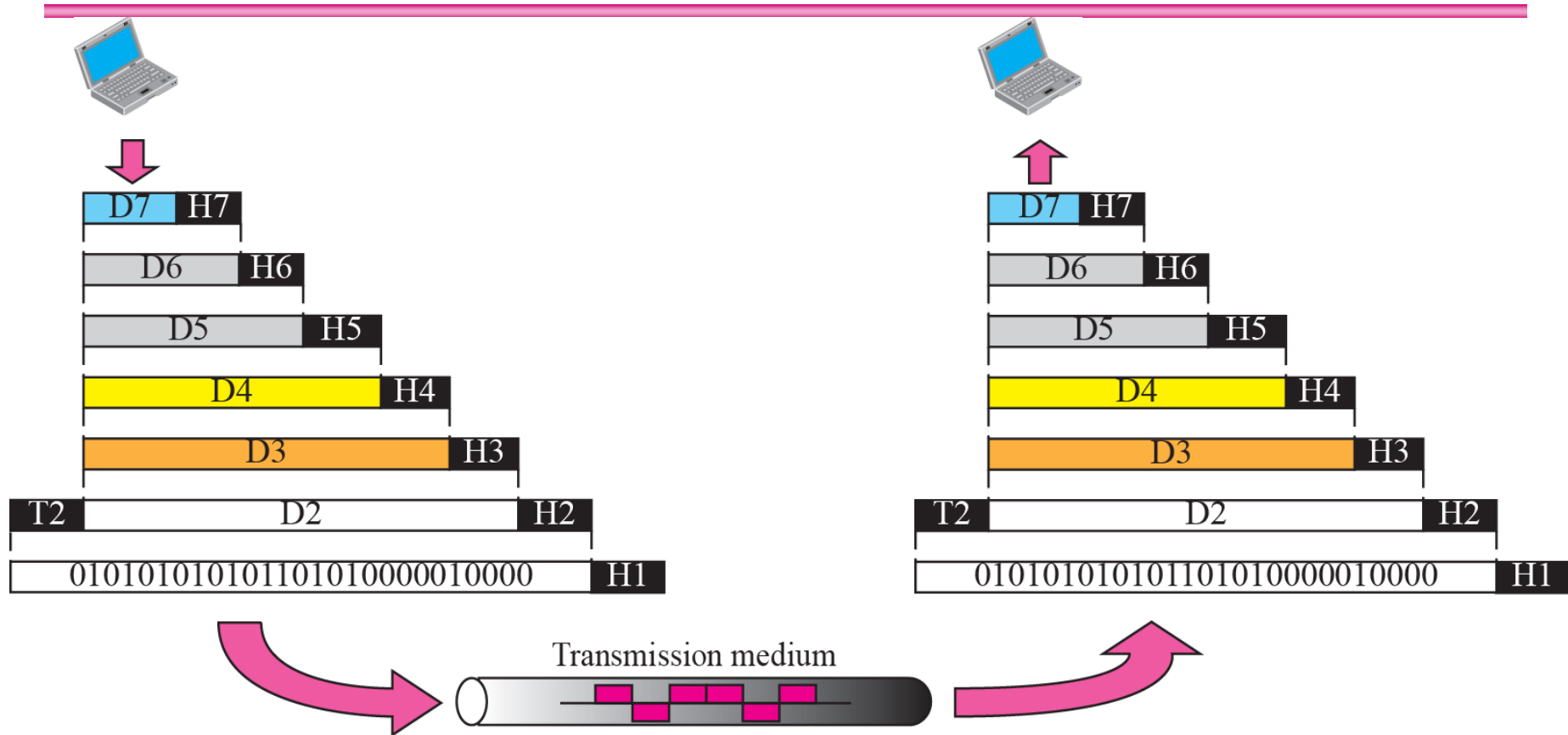


De-encapsulating Data





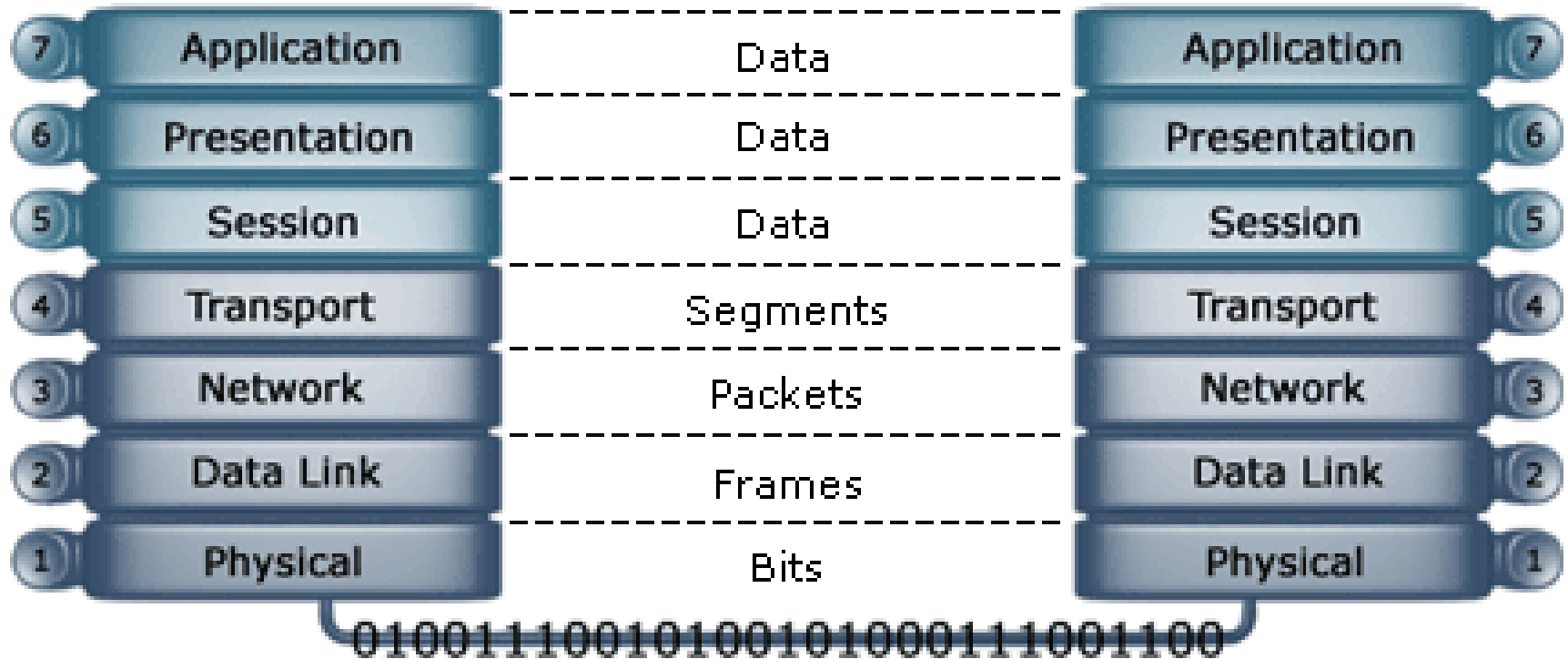
Sending Data



An exchange using the OSI model

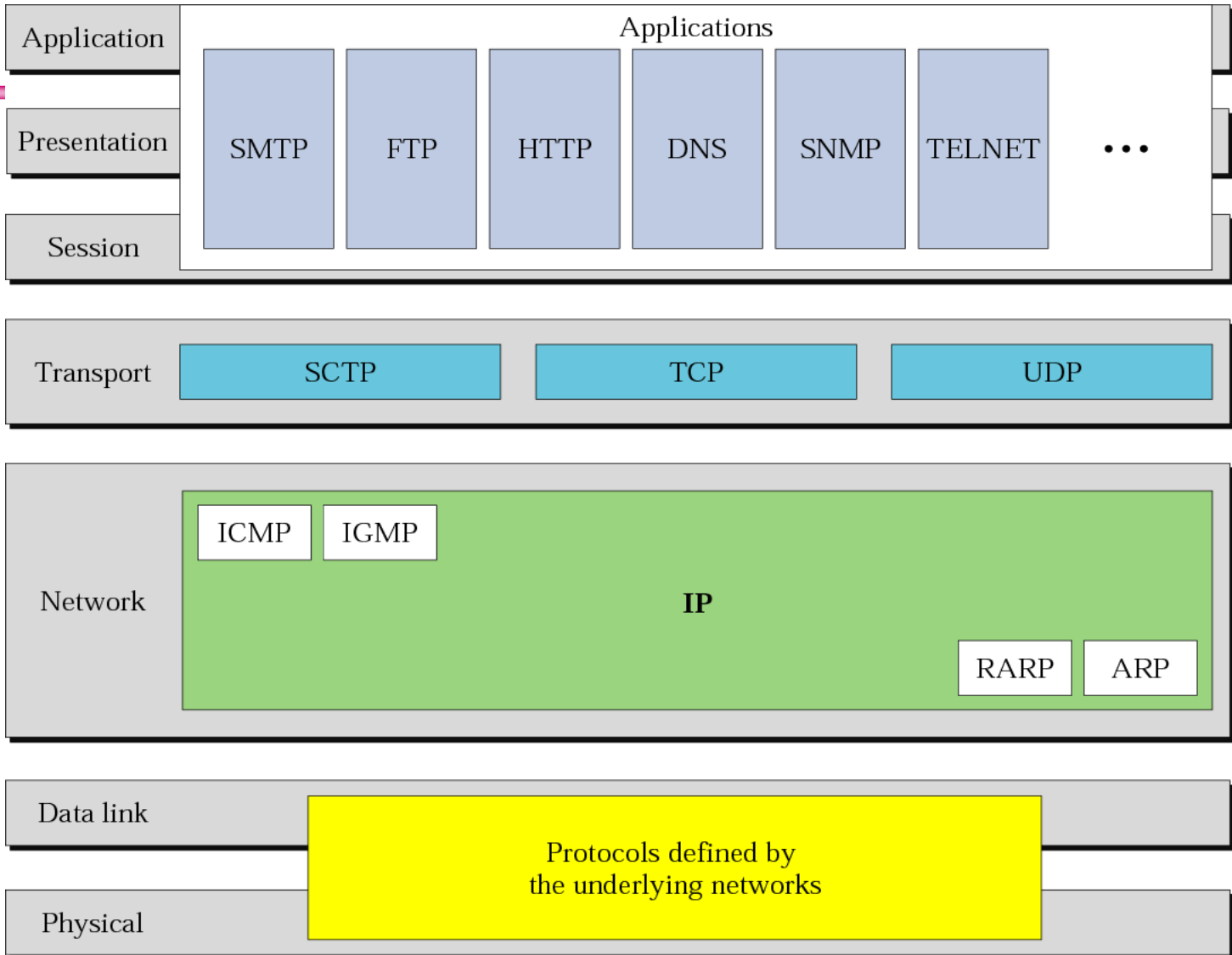


Protocol Data Unit PDU





Protocols on each layers





Summary



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1. How to connect to the internet ?
 2. Basic network devices
 3. Characteristics of network
 4. Network Classification
 5. What is OSI model
 6. Functions of each layer
 7. PDU
 8. Flow of data